

3.1.2 Skill acquisition

This section focuses on how skill is acquired and the impact of psychological factors on performance. Students should develop knowledge and understanding of the principles required to optimise learning of new, and the development of existing, skills in a range of physical activities.

Students should be able to understand and interpret graphical representations associated with skill acquisition theories.

3.1.2.1 Skill, skill continuums and transfer of skills

Content	Additional information
Characteristics of skill.	
Use of skill continua.	Open – closed. Discrete – serial – continuous. Gross – fine. Self-paced – externally paced. High – low. Simple – complex.
Justification of skill placement on each of the continua.	
Transfer of learning.	Positive. Negative. Zero. Bilateral.
Understanding of how transfer of learning impacts on skill development.	

3.1.2.2 Impact of skill classification on structure of practice for learning

Content	Additional information
Methods of presenting practice.	Whole. Progressive part. Whole–part–whole.
Types of practice.	Massed. Distributed. Variable. Mental practice.
Understanding how knowledge of skill classification informs practice structure (presentation and type) to allow learning/ development of skills.	

3.1.2.3 Principles and theories of learning and performance

Content	Additional information
Stages of learning and how feedback differs between the different stages of learning.	Cognitive, associative, autonomous.
Learning plateau.	Causes and solutions.
Cognitive theories.	Insight learning (Gestalt).
Behaviourism.	Operant conditioning (Skinner).
Social learning.	Observational learning (Bandura).
Constructivism.	Social development theory (Vygotsky).
Understanding of how theories of learning impact on skill development.	

3.1.2.4 Use of guidance and feedback

Content	Additional information
Methods of guidance.	Verbal. Visual. Manual. Mechanical.
Understand the different purposes and types of feedback.	Knowledge of performance. Knowledge of results. Positive and negative. Intrinsic. Extrinsic.
Understanding of how feedback and guidance impacts on skill development.	

3.1.2.5 Memory models

3.1.2.5.1 General information processing model, to include:

Content	Additional information
Input.	Senses. Receptors. Proprioception. Perception. DCR process Selective attention.
Decision making.	Short and long term memory
Baddeley and Hitch, working memory model memory system.	Functions and characteristics of components of working memory model.
Output.	
Feedback.	

3.1.2.5.2 Efficiency of information processing to include:

Content	Additional information
Application of Whiting's information processing model to a range of sporting contexts.	
Applied understanding of information processing terms within a sporting context.	Environment. Display. Sensory organs. Perceptual mechanism. Translatory mechanism. Effector mechanism. Muscular system output data. Feedback data.
Definitions of and the relationship between reaction time, response time, movement time.	Simple reaction time. Choice reaction time.
Factors affecting response time.	Hick's law. Psychological refractory period. Single channel hypothesis.

Content	Additional information
Definitions of anticipation.	Temporal. Spatial.
Strategies to improve response time.	
Schmidt's schema theory.	Recall. Recognition. Initial conditions. Response specifications. Sensory consequences. Response outcomes.
Application of schema theory in sporting situations.	
Strategies to improve information processing.	Input – selective attention Decision making process – chunking, chaining, response time, schema.